

mARker: Hybrid-Interfaced Spatial Sketching via iPad AR, Apple Pencil, and an Instantly Crafted Tracker

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ABSTRACT

Mobile AR enables wider access to spatial sketching, yet current research designs rely on crowded on-screen controls, constant device motion, or extra hardware. To address this, we present mARker, a proof-of-concept on the iPad + Apple Pencil Pro ecosystem. It attains reliable stylus pose with a low-cost, self-fabricated cube and uses developer guideline-aligned Pencil gestures for writing-hand intuitive input, while AR session affords natural proxy-plane access and scene-aware placement. Blending spatial and on-screen interaction keeps setup light but rich in functionality. These choices reduce operation overhead and hardware burden, broadening the reach and expressiveness of MAR spatial sketching.

Index Terms: Mixed / augmented reality, Ubiquitous and mobile computing systems and tools.

1 INTRODUCTION

Spatial sketching is widely explored in augmented reality (AR), especially in head-mounted display (HMD) ecosystems; by contrast, mobile AR (MAR) on phones and tablets is less considered, since on-screen interaction and narrow fields of view make simultaneous sketching and viewing awkward. Yet tablets—especially iPad with Apple Pencil—are ubiquitous creative tools; leveraging this ecosystem, a general-purpose MAR spatial sketching tool can meaningfully broaden access and practice.

Prior studies that address spatial sketching in MAR cluster into two approaches: screen-centric methods that rely on device motion/touch and helper interface UIs [7, 3], and spatial-tracking methods that use hand/gesture tracking or custom pens with marker-based pose [4, 5]. Both above approaches come with their own interaction advantages and drawbacks: relying solely on the screen is simple and easy to set up, but the sketching operation depends heavily on moving the device in space and on additional UI tools; relying on spatial tracking leverages the user’s intuitive hand movements, yet the tracking methods are inconsistent and impose prohibitively high hardware setup barriers.

In response, we want to pursue a design that synthesizes complementary strengths, realizing a prototype with (1) ubiquitous setup capability and (2) a natural, intuitive interface covering representative functions. We then introduce mARker: it addresses these gaps by building on the standardized iPad + Apple Pencil ecosystem with only a lightweight tracking cube, leveraging Pencil input APIs and iPad AR to deliver core sketching with Apple Human Interface Guideline (HIG)–aligned interactions [1], while minimizing screen controls and setup friction.

2 CANVAS SETUP AND USER INTERFACES

The official components for mARker are (1) an Apple Pencil Pro, (2) a compatible iPad, and (3) a tracking cube. mARker’s AR app

is built under iPadOS 17.5 and supports two interfaces (Fig. 1): a spatial interface for mid-air sketching via stylus tracking, and an on-screen interface for proxy plane sketching, entity transformation, along with auxiliary controls such as color selection. The tracking cube is custom-designed for the Apple Pencil and can be fabricated within an hour at a material cost of less than one U.S. dollar, with 3D printed parts, trimmed paper prints, and a magnet.

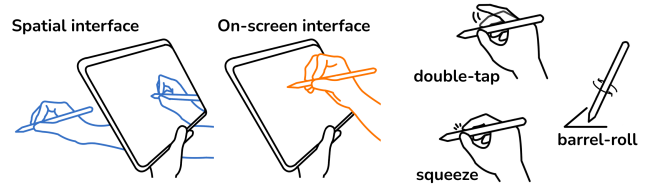


Figure 1: Schematics of two sketching interfaces and Apple Pencil Pro input gestures employed by mARker.

Inputs with Apple Pencil The development of mARker benefits greatly from the new APIs introduced at WWDC 24 for Apple Pencil Pro [1]. The squeeze gesture can be invoked when the Pencil is not touching the screen, and the barrel-roll gesture lets users rotate elements with a natural rotation gesture. In mARker, we employ the double-tap, squeeze, barrel-roll APIs to create our interactive interfaces, and their specific on-Pencil gestures are also shown in Fig. 1. For demonstration purposes, we implemented two sketching modes: freehand and ruler. double-tap serves as a global toggle between these modes, while squeeze and barrel-roll are assigned to interface-specific functions that will be detailed in later sections.

Spatial interface The spatial interface is designed to track the Pencil in real-time and let users apply on-Pencil gestures to execute strokes in the AR scene.

We implement multi-fiducial marker tracking to capture the stylus’s 6-DoF pose, a method widely used in passive stylus work [5, 6, 2]. Markers are rigidly mounted via a magnetized collar-cube that snaps to a normal-up orientation using the Pencil’s built-in magnets (Fig. 2 (a)), preserving grip and maintaining a fixed rigid transform with respect to the Pencil; multiple faces improve robustness during motion and occlusion. We estimate per-marker poses using the OpenCV framework, fuse them through the fixed mounting geometry, and stabilize with short-window and One-Euro filtering. A virtual tip is rendered at the end of the tracking pipeline for visual feedback and stroke execution.

The squeeze gesture toggles stroke execution in the spatial interface, aligning with Apple HIG’s guidance to *treat squeeze as a single, quick gesture that performs a discrete action*. As shown in Fig. 2 (b), this enables one-handed control of pen down/up without touching on-screen UI.

On-screen interface The on-screen interface lets users apply screen scribble to execute strokes as an alternative option, and also

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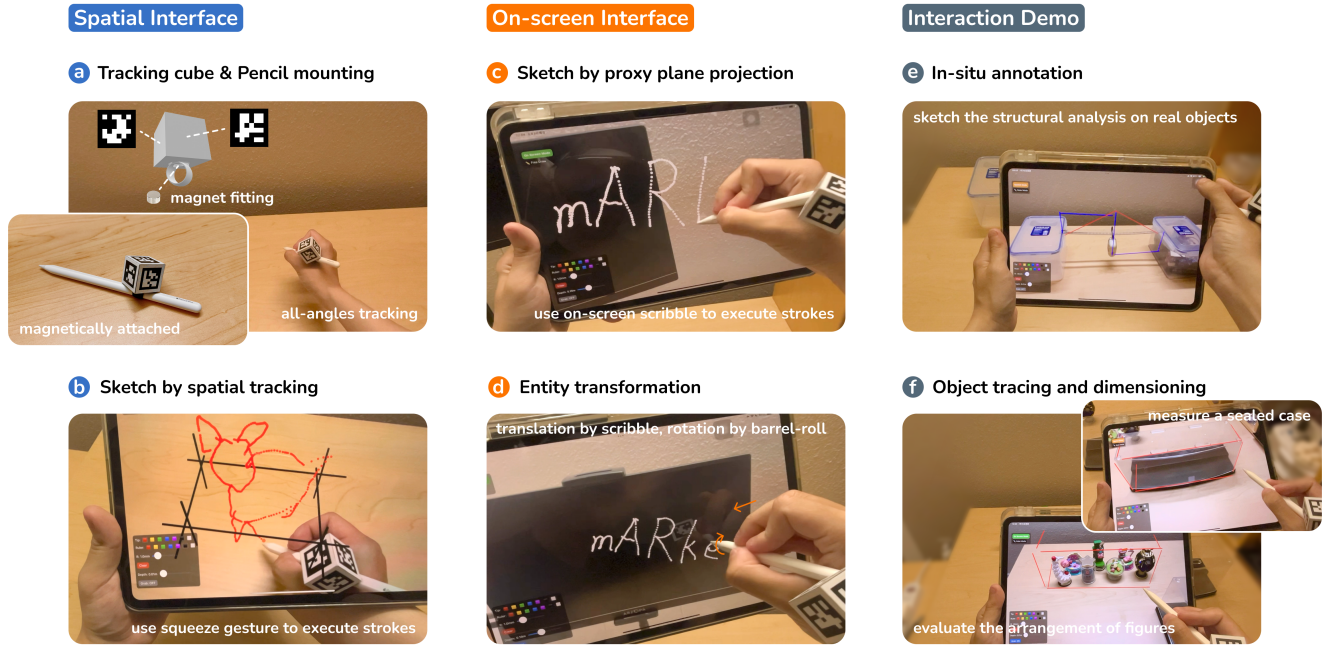


Figure 2: Implementation & functionality details of each individual interface, and additional sketching scenarios using mARKer. Spatial interface: (a) tracking cube design and mounting the cube on Apple Pencil, (b) sketching in mid-air through tracking and the squeeze gesture. On-screen interface: (c) sketching on screen with projection into AR scene, (d) moving sketches with Pencil gestures. Interaction demo: (e) in-situ analysis diagram annotation on real object, (f) dimensioning an object by physically tracing it and evaluating the dimension.

use it to manage strokes as a grab tool.

Using the AR session’s camera pose, we define a proxy plane parallel to the iPad display and project on-screen scribbles onto it, rendering and recording strokes at the corresponding scene depth (Fig. 2 (c)). Users can reposition the plane by moving the iPad and toggle depth to draw at different locations. Per HIG—*Let people make a mark the moment Apple Pencil touches the screen*—screen contact activates stroke execution, distinct from the squeeze-based control used in the spatial interface.

In grab mode (Fig. 2 (d)), Pencil gestures manipulate existing strokes: the on-screen trajectory translates entities within their current depth plane, while barrel-roll rotates them around the camera’s forward axis. We map Pencil roll angle change to rotation for direct control, consistent with HIG guidance to *use barrel roll only to modify marking behavior*.

3 INTERACTION DEMONSTRATION

We envision mARKer as a higher-dimensional sketching tool that supports diverse scenarios with deeper real-world interactivity. Beyond the previously demonstrated 3D painting and scene decoration, additional scenarios are expectable. For example, mARKer can be used to directly annotate objects in the environment, giving an in-situ demonstration instead of traditional on-paper analysis diagrams (Fig. 2 (e)). Alternatively, users can use the Pencil to touch and trace objects to capture measurements, and then use those measurements to understand how the object physically fits within its surroundings (Fig. 2 (f)).

Live demo The live demo aims to showcase mARKer’s low-overhead, ubiquitous setup, and also spatial sketching functionalities: after a brief configuration—or on a preconfigured device—attendees can immediately explore spatial sketching of their preference. During the session, we’ll provide materials and also preassembled units of the tracking cube, along with an iPad with Apple Pencil Pro. Participants may also bring their own iPads to

complete the setup by just downloading the App. As a general-purpose sketching tool, we do not impose any prescribed tasks for the demo: free drawing as you wish.

4 CONCLUSION AND FUTURE WORKS

Built on standardized tablet hardware and a hybrid stylus interface, mARKer is a proof-of-concept ubiquitous spatial sketching tool that runs on commodity iPad + Apple Pencil with a self-fabricatable tracking cube. Leveraging Pencil APIs and HIG-aligned behaviors, it minimizes on-screen UI and extends familiar 2D screen workflows along with spatial functions: users can access drawing through screen-proxy-plane or direct spatial manipulation, and employ Pencil-coupled gestures for natural stroke execution & entity transformation. Future works can focus on refining pose tracking (better fiducial methods and potential machine learning models to reduce marker reliance), improving AR rendering, and evolving toward a vector-graphics workspace for high-quality 3D authoring.

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